

Chapter 62: Connective Tissue Disorders — Revision Notes

Williams Obstetrics 26e, pp. 2888–2932 of the PDF. High-yield numbers are in **bold**. The seven tables are transcribed from the PDF images (they are missing from the extracted chapter text). This chapter has no figures.

Big picture: Connective tissue disorders are either **immune-complex (rheumatic) diseases** — SLE, RA, scleroderma, vasculitis, myopathies — or **inherited structural defects** — Marfan, Ehlers-Danlos, osteogenesis imperfecta. The immunosuppression of pregnancy (Th2 shift, progesterone) quiets some diseases (RA), while others flare unpredictably (SLE). The single most important predictor of a good outcome is **quiescent disease for ≥6 months before conception**. **APS** is the key acquired thrombophilia behind stillbirth, early preeclampsia and FGR.

1. Immune-Mediated Connective Tissue Diseases

- **Two groups:**
 - **With autoantibodies (rheumatoid factor-associated):** SLE, RA, systemic sclerosis, mixed CTD, dermatomyositis, polymyositis, vasculitides.
 - **RF-seronegative spondyloarthropathies (HLA-B27):** ankylosing spondylitis, psoriatic arthritis, Reiter disease.
- **Why pregnancy can mitigate disease:** estrogens upregulate and androgens downregulate T cells; **progesterone is immunosuppressive; Th2 predominance** over Th1.
- **Unrecognized systemic rheumatic disease (~0.3%) → preeclampsia or FGR in 25%.**
- **Fetal microchimerism:** fetal cells and cfDNA persist in maternal blood/organs and may trigger autoantibodies (seen in autoimmune thyroiditis, **systemic sclerosis**, SLE, RA). Engrafted maternal cells may also provoke autoimmunity in offspring.

2. Systemic Lupus Erythematosus (SLE)

Pathogenesis and genetics

- Tissue-binding autoantibodies and immune complexes against **nuclear components** + impaired regulatory T-cell function. Autoantibodies usually precede clinical disease by years.
- **Concordance: monozygotic 25% vs dizygotic 2%; 10%** if one family member affected. Genes: chromosome 16 "autoimmunity gene", HLA-DR3, STAT4, APOL1, FCGR3A, DNASE1.
- A child's HLA-DRB1 genotype can raise maternal SLE risk. Neonatal lupus occurred even after **oocyte donation** to an anti-Ro/La-positive mother (antibody-mediated).

Table 62-1. Some Autoantibodies Produced in Patients with Systemic Lupus Erythematosus

Antibody	Prevalence (%)	Clinical Associations
Antinuclear (ANA)	84–98	Best screening test , multiple antibodies; a second negative test makes SLE unlikely
Anti-dsDNA	62–70	High titers SLE-specific ; may correlate with disease activity, nephritis , and vasculitis
Anti-Sm (Smith)	25–38	Specific for SLE
Anti-SS-A (Ro)	30–49	Not SLE-specific; associated with sicca syndrome, predisposes to cutaneous lupus, neonatal lupus with heart block , reduced risk of nephritis
Anti-SS-B (La)	10–35	Associated with anti-Ro
Antiphospholipid	21–50	Lupus anticoagulant and anticardiolipin antibodies associated with thrombosis, fetal loss, thrombocytopenia , valvular heart disease; false-positive test for syphilis
Anti-RNP	33–40	Not SLE-specific, high titers associated with rheumatic syndromes
Antihistone	70	Common in drug-induced lupus
Antierythrocyte	60	Direct Coombs test, may develop hemolysis
Antiplatelet	30	Thrombocytopenia in 15%; poor clinical test

ANA = antinuclear antibody; dsDNA = double-stranded DNA; RNP = ribonucleoprotein; SLE = systemic lupus erythematosus. Data from Arbuckle, 2003; Hahn, 2018; Petri, 2020a,b.

Clinical manifestations and diagnosis

- **~90% of cases in women**; more common in African-American women. **~1 in 500** women of reproductive age.
- Variable presentation: may start in one organ or be multisystem. Common: malaise, fever, **arthritis, rash, pleuropericarditis, photosensitivity**, anemia, cognitive dysfunction. **≥ Half have renal involvement.**

Table 62-2. Some Clinical Manifestations of Systemic Lupus Erythematosus

Organ System	Clinical Manifestations	Percentage
Systemic	Fatigue, malaise, fever, weight loss	~95
Musculoskeletal	Arthralgias, myalgias, polyarthritis, myopathy	~95
Hematological	Anemia, hemolysis, leukopenia, thrombocytopenia, lupus anticoagulant, splenomegaly	70–85
Cutaneous	Malar (butterfly) rash, discoid rash, photosensitivity, oral ulcers, alopecia, skin rashes	50–80
Neurological	Cognitive dysfunction, mood disorder, headache, seizures, stroke	40–60
Cardiopulmonary	Pleuritis, pericarditis, myocarditis, endocarditis, pneumonitis, pulmonary hypertension	30–60
Renal	Proteinuria, nephrotic syndrome, renal failure	30–50
Gastrointestinal	Nausea, pain, diarrhea, abnormal liver enzyme levels	30–40
Vascular	Thrombosis: venous (10%), arterial (5%)	10–15
Ocular	Conjunctivitis, sicca syndrome	10–15

- **Labs:**
 - **ANA = best screen, but not specific** (low titers in normal people, other autoimmune disease, viral infection, chronic inflammation, drugs).
 - **Anti-dsDNA and anti-Sm are relatively specific.**
 - Anemia, leukopenia, thrombocytopenia; proteinuria and casts with glomerular disease (renal insufficiency commoner with APAs).
 - **False-positive syphilis serology, prolonged PTT, ↑ RF.**
 - Persistent unexplained ↑ **D-dimer** → **high thrombosis risk.**

- **Drug-induced lupus:** PPIs, thiazides, antifungals, antihypertensives, chemotherapy, statins, antiepileptics.
Rarely glomerulonephritis; regresses when the drug stops (antihistone antibodies).

Table 62-3. Clinical Criteria for Classification of Systemic Lupus Erythematosus

Clinical Manifestations	Immunological Manifestations
Skin	ANA
Oral ulcers	Anti-dsDNA
Alopecia	Anti-Smith
Synovitis	Antiphospholipid
Serositis	Hypocomplementemia
Renal: proteinuria, casts, biopsy	Direct Coombs
Neurological: seizures, psychosis, myelitis, neuropathies, confusional state	
Hemolytic anemia	
Leukopenia	
Lymphopenia	
Thrombocytopenia	

ANA = antinuclear antibodies; dsDNA = double-stranded DNA; Sm = Smith. Data from Hahn, 2018; Petri, 2020a,b.

- **Diagnosis: ≥ 4 criteria, serially or simultaneously.**

The text says "four or more of the 11 clinical criteria"; the table lists 11 clinical plus 6 immunological items. In the SLICC scheme on which it is based, ≥ 4 criteria are needed including at least one clinical and one immunological item (or biopsy-proven lupus nephritis with ANA or anti-dsDNA).

3. Lupus and Pregnancy

- **~1 in 900 births** (meta-analysis of ~9 million pregnancies).
- **Rule of thirds: improves in 1/3, unchanged in 1/3, worsens in 1/3** — flares can occur without warning.
- **Maternal mortality 325 per 100,000** (13,555 pregnancies); deaths occur almost only with **active disease**.
- ↑ Thrombosis, infection, PPH, transfusion.

Table 62-4. Risk of Adverse Pregnancy Outcomes in Women with Systemic Lupus Erythematosus

Outcome	Relative Risk (CI)
Preeclampsia/Eclampsia	3.4 (3.2–3.6)
Cesarean delivery	1.4 (1.1–1.7)
Stillbirth	16.5 (2.9–92.1)
Fetal loss	7.6 (4.8–11.9)
Preterm birth	2.3 (1.8–3.1)
SGA infants	2.5 (1.4–4.5)
LBW infants	4.8 (3.7–6.3)
NICU admission	2.8 (2.3–3.4)

CI = confidence interval; LBW = low birth weight; NICU = neonatal intensive care unit; SGA = small for gestational age.

- Outcomes much improved; favorable for inactive or mild/moderate disease. **Confined cutaneous lupus: usually no adverse outcome. SLE newly diagnosed in pregnancy tends to be severe.**

- **Best outcome when:**
 1. **Lupus quiescent ≥ 6 months before conception**
 2. **No lupus nephritis** (proteinuria or renal dysfunction)
 3. **No APS or lupus anticoagulant**
 4. **No superimposed preeclampsia**

Lupus nephritis

- Active nephritis → adverse outcomes (higher histological class → higher risk); much better if in remission.
- Preeclampsia **27% with pre-existing renal disease vs 16% without.**
- 309 pregnancies: **flare 30%** (40% of these with renal insufficiency); **maternal mortality 1.3%**; $\frac{1}{3}$ preterm (later cohort).
- **Continue immunosuppression.** Permissible: **hydroxychloroquine, cyclosporine, azathioprine, tacrolimus.**
- **New or severe renal flare → tacrolimus** $\pm$ IV corticosteroids, other immunosuppressants, IVIG.

Lupus flare vs preeclampsia

- **Chronic hypertension in up to 30%** of SLE pregnancies; preeclampsia common, **earlier and more frequent with nephritis or APAs.**
- Both: hypertension, proteinuria, edema, worsening renal function. **Management differs: flare → immunosuppression; severe preeclampsia → delivery.**
- May be impossible to separate if the kidney is the only organ involved. Helpful tests: **uric acid, complement (C3/C4), dsDNA titers.** CNS lupus can cause **seizures mimicking eclampsia.**

Feature	Lupus nephritis flare	Preeclampsia
Complement (C3, C4)	Falling/low	Normal ($\uparrow$ in pregnancy)
Anti-dsDNA	Rising	Unchanged
Uric acid	Usually normal	Often $\uparrow$
Other organs	Rash, arthritis, serositis, active sediment	HELLP features, headache, visual symptoms
Treatment	Immunosuppression	Delivery

Pearl: active urinary sediment (RBC casts) favors lupus nephritis; rising transaminases with low platelets favor HELLP.

Management during pregnancy

- **Baseline:** 24-h urine protein, serum creatinine, CBC → repeat **each trimester** (monthly if symptoms).
- **C3, C4 and dsDNA** at the first visit and with symptoms. Complement normally **rises** in pregnancy; low/falling C3/C4 suggests activity, but normal/high levels don't exclude a flare. **Falling complement + rising dsDNA with symptoms = flare.**
- **Screen for antiphospholipid antibodies** and **anti-SS-A (Ro) / anti-SS-B (La)** at the start of pregnancy.
- SLE hemolysis: Coombs+, anemia, reticulocytosis, unconjugated hyperbilirubinemia. **Early chronic thrombocytopenia → APAs**; later → preeclampsia.
- **Fetus: sonography every 4 wk from 24 wk** (growth, oligohydramnios); **antepartum testing in the 3rd trimester.** Allow to reach term unless compromise.

Pharmacological treatment

Drug	Pregnancy use
NSAIDs	Arthralgia/serositis; avoid chronic or large intermittent doses (oligohydramnios, ductus closure)
Hydroxychloroquine	Continue (stopping precipitates flares); no malformations; start in 1st trimester — ↓ SGA 85% ; ↓ congenital heart block 50% with anti-SSA; only treatment that ↓ neonatal lupus
Low-dose aspirin	Throughout gestation
Corticosteroids	Severe disease: prednisone 1–2 mg/kg/d or methylprednisolone 1000 mg IV over 90 min daily × 3 d , then taper to 10–15 mg every morning ; risk of GDM
Stress-dose steroids	Chronic/recent steroid users in labor: hydrocortisone 100 mg IV q8h × 48 h
Azathioprine	Good safety record; 2 mg/kg/d
Tacrolimus, rituximab, IVIG	Nephritis or steroid-resistant disease
✗ Mycophenolate, methotrexate, cyclophosphamide	Teratogenic — avoid (cyclophosphamide can be considered in 2nd/3rd trimester for severe disease)
✗ ACE inhibitors / ARBs	Switch to nifedipine/CCB, methyldopa or labetalol

- No cure; complete remission rare. **~¼ have mild disease** (painful, fatiguing but not life-threatening).

Perinatal outcome

- ↑ Preterm birth, FGR, stillbirth, neonatal lupus. Worse with active disease, significant proteinuria, renal impairment, chronic hypertension, preeclampsia, **neuropsychiatric lupus**.
- Mechanism: **decidual vasculopathy with placental infarction** and ↓ perfusion.

Neonatal lupus syndrome

- **Lupus dermatitis**, hematological/systemic abnormalities, occasionally **congenital heart block**. Due to **anti-SS-A and SS-B** (present in **up to 50%** of women with SLE).
- **Skin lesions in 30–40%**, appearing at **4–6 weeks**; thrombocytopenia and hepatic involvement in 5–10%. (128 infants: cutaneous 16%, hematological/transaminase abnormalities 26%.)
- **Skin, thrombocytopenia and hemolysis are transient** (clear in months).
- **Recurrence in later children up to 25%. Hydroxychloroquine is the only maternal treatment that ↓ risk.**

Congenital heart block

- **Myocarditis and fibrosis between the AV node and bundle of His**; almost exclusively with **anti-SS-A/SS-B**. Also with **Sjögren syndrome**.
- **Risk 2–3% with antibodies → 20% if a prior child was affected.**
- **Permanent**; pacemaker usually needed; poor prognosis — **~20% of 325 affected infants died** (⅓ of these stillborn).
- **Maternal corticosteroids, plasma exchange or IVIG do NOT prevent it.** Hydroxychloroquine is associated with ↓ incidence (especially after a prior affected child).
- **Fetal cardiac monitoring 18–26 wk** in antibody-positive women.
- Steroid treatment of established block is controversial (no RCTs). Observational: no block in 26 neonates exposed to maternal steroids **before 16 wk**, vs 15 of 61 when steroids started after 16 wk.

Memory hook: "Ro/La → Rash at 4–6 weeks (transient) and Rhythm block (permanent)." Heart block risk **2–3%, 20%** after an affected sibling.

Long-term prognosis and contraception

- **Survival: 95% at 5 yr, 90% at 10 yr, 78% at 20 yr.** Deaths: infection, flares, end-organ failure, hypertension, stroke, CVD.
- **Combined OCs did not ↑ flares** (two large trials) — ⚠️ but **APAs contraindicate estrogen**. Progestin implants/injections fine; **IUDs acceptable** (no ↑ infection with immunosuppression). Sterilization safest postpartum or when quiescent.

4. Antiphospholipid Syndrome (APS)

Definition and features

- **Autoantibody-mediated acquired thrombophilia:** persistent APAs **plus** arterial/venous thrombosis **or** obstetric morbidity.
- **APAs: lupus anticoagulant (LAC), anticardiolipin (ACA), anti-β2 glycoprotein-I;** IgG, IgM or IgA. Most common with SLE, other CTDs and APS; low levels in some normal people.
- **Mechanisms:** (1) procoagulant activation, (2) inactivation of natural anticoagulants, (3) **complement activation**, (4) **inhibition of syncytiotrophoblast differentiation**.
- Any organ: **CNS prominent** (arterial/venous stroke, psychiatric features, MS-like); renovascular failure (hard to separate from lupus nephritis); peripheral/visceral thrombosis (e.g., mesenteric infarction).
- **Obstetric:** recurrent loss, FGR, **stillbirth, early-onset preeclampsia**, preterm birth. Treatment (aspirin + heparin + monitoring) → **live births >70%**.
- **Catastrophic APS (Asherson syndrome):** rapidly progressive thrombosis in **≥3 organs/tissues** simultaneously; cytokine storm, high mortality; trigger found in half.

Specific antibodies

1. **Anti-β2 glycoprotein-I:** β2GP-I (apolipoprotein H) normally inhibits platelet prothrombinase and aggregation; it is **highly expressed on syncytiotrophoblast** → antibodies promote thrombosis at the placental interface.
2. **Lupus anticoagulant:** prolongs PT, PTT and **Russell viper venom time in vitro**, but is **thrombotic in vivo**. **The only APA consistently associated with adverse pregnancy outcomes.**
3. **Anticardiolipin:** against cardiolipin in mitochondrial membranes and platelets.
 - Antibodies against **protein C, protein S, annexin V** (syncytiotrophoblast) exist — **testing not recommended** (nonconventional antibodies may matter in seronegative APS).

Diagnosis (international consensus; Table 11-4)

- **≥1 clinical criterion** (vascular thrombosis or specified pregnancy morbidity) **+ laboratory criterion confirmed twice, 12 weeks apart:** medium/high-titer IgG or IgM **ACA** or **anti-β2GP-I**, or **LAC** positivity.
- **LAC tests:** PTT usually prolonged; **dilute Russell viper venom test more specific**. Only **~20%** of APS patients have LAC alone → also do **ACA ELISA**. Assays poorly standardized; large interlaboratory variation.

Table 62-5. Some Clinical Features of Antiphospholipid Syndrome

Category	Features
Venous thrombosis	Thromboembolism, thrombophlebitis, livedo reticularis
Arterial thrombosis	Stroke, transient ischemic attack, Libman-Sacks cardiac vegetations , myocardial ischemia, distal extremity and visceral thrombosis and gangrene
Hematological	Thrombocytopenia, autoimmune hemolytic anemia
Renal	Thrombotic microangiopathy, glomerular thrombosis
Other	Neurological manifestations, migraine headaches, epilepsy; renal artery or vein thrombosis; arthritis and arthralgia
Pregnancy	Preeclampsia and HELLP syndrome, recurrent miscarriage, preterm delivery, fetal-growth restriction, fetal death

HELLP = hemolysis, elevated liver enzyme levels, low platelet count. Data from Garcia, 2018; Giannakopoulos, 2013; Moutsopoulos, 2018.

Pregnancy and APAs

- **Up to 30% of SLE patients have APAs.** APA positivity: **~5%** of the general obstetric population; **7%** of women without autoimmune disease who have pregnancy complications.
- With APAs (especially LAC): ↑ decidual vasculopathy, placental infarction, **FGR, early-onset preeclampsia, recurrent fetal death**; also venous/arterial thrombosis, cerebral thrombosis, hemolysis, thrombocytopenia, pulmonary hypertension.
- **ACA + anti-β2GP-I together → more miscarriage** than either alone; **higher titers → worse outcomes.**
- **Pathophysiology:** platelet damage/aggregation; APA injury to endothelial/syncytiotrophoblast membranes via β2GP-I or **annexin V** → exposed basement membrane → thrombus; ↓ decidual PGE₂; ↓ protein C/S activity; **tissue factor pathway activation; placental complement activation.** Not explained by thrombosis alone — **inflammation "second hit"**.

Adverse outcomes

- **Fetal deaths are more characteristic of APS than first-trimester miscarriage.**
- Stillbirth case-control (582 vs 1547): **3–5× stillbirth** with ↑ ACA and anti-β2GP-I.
- **Worse outcomes with:** (1) **all three APAs (triple positive)**, (2) comorbid SLE/autoimmune disease, (3) **prior thrombosis and pregnancy morbidity.**
- Pregnancy failure probability **93% with ≥2 APA types vs 63% with one.**

Thrombosis prevention

- APA titers reported as GPL/MPL/APL units: negative, low, medium, high positive.
- **Prior thrombosis + APAs → prophylactic anticoagulation (heparin) throughout pregnancy and 6 weeks postpartum** (ACOG).
- **No prior thrombosis:** options vary (Table 55-5) — close observation ± prophylactic or intermediate-dose heparin, then **postpartum anticoagulation for 6 weeks.**
 - Some treat if **medium/high-titer ACA or LAC + unexplained prior 2nd/3rd-trimester fetal death**, or recurrent early loss with medium/high titers.
- **Serial growth scans + 3rd-trimester antepartum testing.**

Specific therapy

- **Low-dose aspirin 60–80 mg/d:** blocks thromboxane A₂ while sparing prostacyclin. Only side effect: slight surgical small-vessel bleeding. **Doesn't help APAs without full APS;** recommended for **SLE or APS.**

- **Heparin:**
 - **UFH 5000–10,000 units SC q12h** or **enoxaparin 40 mg daily**.
 - With therapeutic dosing, **measure heparin (anti-Xa) levels** — LAC distorts clotting tests.
 - Prevents venous/arterial and **microcirculatory (decidual–trophoblastic)** thrombosis; binds β 2GP-I on syncytiotrophoblast, blocking antibody binding; binds APAs.
 - Complications: bleeding, thrombocytopenia, **osteopenia/osteoporosis**.
- **Aspirin + heparin = most effective regimen**, yet **~30% still fail**.
- **Corticosteroids: NOT for primary APS**; used for SLE/secondary APS at the lowest effective dose.
- **IVIG:** controversial; for CAPS, HIT, or failed first-line therapy with preeclampsia/FGR. **0.4 g/kg/d × 5 d (2 g/kg) monthly, or 1 g/kg monthly**; >\$10,000/course; no benefit added to aspirin + LMWH; no $\uparrow$ live birth in recurrent loss (Cochrane).
- **Hydroxychloroquine + aspirin** (SLE): may $\downarrow$ thrombosis and improve outcomes in APS (investigational for APS alone).
- **Pravastatin** added to aspirin + LMWH improved placental flow and outcomes in 21 women with FGR/preeclampsia — needs larger trials.
- **CAPS: full anticoagulation + high-dose corticosteroids + plasma exchange and/or IVIG.**

Treatment efficacy

- Untreated APS (especially LAC+) $\rightarrow$ frequent fetal loss; **even treated, recurrent fetal loss 20–30%**.
- Some women with SLE + APAs, even LAC with prior losses, have live births untreated.

Table 62-6. Pregnancy Outcomes (%) in 750 Women Treated for Antiphospholipid Syndrome—the PREGNANTS Study

Outcome	Triple-Positive (n = 20)	Double-Positive LAC Negative (n = 90)	LAC Alone (n = 54)	ACA Alone (n = 458)	Anti- β 2 Glycoprotein-I (n = 128)
Livebirth	30	43	80	56	48
Stillbirth	45	34	7	21	30
Preeclampsia ^a	55	54	11	34	48

^aNonsevere only. ACA = anticardiolipin antibodies; LAC = lupus anticoagulant. All women received low-dose aspirin + prophylactic LMWH from the 1st trimester (Saccone, 2017).

Memory hook: "Triple positive = triple trouble" — only **30%** live births despite aspirin + LMWH, with **45%** stillbirth and **55%** preeclampsia.

Neonatal effects

- **Up to 30% of newborns** have passively acquired APAs.
- Reported: $\uparrow$ learning disabilities, **4× perinatal stroke**; European registry (141): preterm 16%, LBW 17%, later behavioral problems 4%, **no neonatal thrombosis**; autism in 3 of 36 (all with persistent neonatal anti- β 2GP-I IgG).

5. Rheumatoid Arthritis (RA)

Features

- T-cell cytokine-driven **inflammatory synovitis of peripheral joints** $\rightarrow$ cartilage destruction, erosions, deformity; pain worse with movement.
- Extraarticular: **rheumatoid nodules, vasculitis, pleuropulmonary disease**; fatigue, anorexia, depression.

- **Prevalence 0.5–1%; F:M = 3:1; peak onset 25–55 yr.** Heritability 15–30%; >30 loci; **HLA-DR4, HLA-DRB1.**
- **Pregnancy protects against developing RA** (HLA-disparate fetal microchimerism).

Management

- Goals: pain relief, ↓ inflammation, joint protection, function (physical/occupational therapy).
- **NSAIDs** (COX-1 and COX-2): in pregnancy — small ↑ **cardiac malformations** (1st trimester), early miscarriage, **ductus arteriosus constriction, neonatal pulmonary hypertension**; also gastritis/bleeding. They **don't slow disease**.
- **Glucocorticoids: prednisone 7.5 mg/d** for the first 2 yr of active disease ↓ erosions. **Flares: sulfasalazine and hydroxychloroquine (safe) + prednisone 7.5–20 mg/d.**
- **DMARDs: methotrexate** is preferred outside pregnancy — **✗ methotrexate and leflunomide are contraindicated in pregnancy.** ¼ of women with RA took a DMARD within 6 months of conception; 2.9% had unplanned methotrexate exposure.
- **Biologic DMARDs (TNF-α inhibitors):** infliximab, adalimumab, golimumab, certolizumab, etanercept — limited data but **appear safe** (300 exposures, no fetal effects; adalimumab 74 women no risk). Little known about anakinra, rituximab, abatacept, tocilizumab, tofacitinib.

Pregnancy and RA

- **Improves in up to 70%** (regulatory T cells, pregnancy-associated α2-glycoprotein, **maternal–fetal HLA class II disparity**).
- **⚠ Postpartum flare in 40–50%.** New-onset RA risk **6× in the first 3 months postpartum**; breastfeeding >12 months ↓ later RA.
- **Fertility ↓**; ↑ hypertensive disease, preterm birth, hemorrhage, FGR (~32,000 women).
- **Cervical spine involvement:** subluxation risk (joint laxity) → **⚠ intubation concerns.**

Juvenile rheumatoid arthritis

- Most common chronic arthritis of childhood; persists into adulthood.
- Pregnancy: usually quiescent; **postpartum flares common.** Joint deformities: **15 of 20 cesareans for contracted pelvis or joint prostheses.**
- ↑ Preterm birth (± FGR, preeclampsia); fetal development normal. Early disease severity predicts preterm birth and FGR; low 1st-trimester activity predicts low 3rd-trimester activity.

6. Systemic Sclerosis (Scleroderma)

- Microvascular damage + immune activation + **excess collagen** in skin, lungs, heart, GI tract, kidneys. **F:M = 5:1; age 30–50.**
- **Microchimerism:** Y-chromosome DNA in almost half of affected women vs 4% of controls; incidence ↑ in years after childbirth.

Clinical course

Form	Features
Limited cutaneous (e.g., CREST : Calcinosis, Raynaud, Esophageal dysmotility, Sclerodactyly, Telangiectasia)	More benign, slow
Diffuse cutaneous	Rapid skin fibrosis → GI fibrosis (distal esophagus), lungs, kidneys

- **ANA in 95%; Raynaud phenomenon in 95%;** esophageal symptoms (fullness, epigastric burning) in half; **pulmonary hypertension in 15%.**
- **10-yr survival 70% with pulmonary fibrosis; pulmonary arterial hypertension = main cause of death.**

- **Scleroderma renal crisis** (obliterative renal cortical vasculopathy) in **up to 1/4** → renal failure + malignant hypertension. ACE inhibitors may be needed despite teratogenicity.
- Pulmonary hypertension → bosentan or sildenafil.
- **Overlap syndrome; mixed CTD** (SLE + SSc + polymyositis + RA features, **high anti-RNP**) = undifferentiated CTD.

Pregnancy

- **~1 in 22,000 pregnancies.** Stable if baseline function good.
- **Dysphagia and reflux worsen** (loss of peristalsis in the lower 2/3 of the esophagus).
- Renal insufficiency/malignant hypertension → ↑ superimposed preeclampsia. **Rapidly worsening renal or cardiac disease → consider termination.**
- **⚠ Renal crisis: uncommon but life-threatening, and does NOT improve with delivery.**
- **Pulmonary hypertension usually contraindicates pregnancy.**
- **Delivery:** vaginal unless soft-tissue thickening causes dystocia. **Difficult intubation (small mouth opening) + aspiration risk → neuraxial analgesia preferred.** Warm room, fluids, blankets, socks and gloves (Raynaud). Stress-dose steroids if used.
- **Outcomes:** ↑ preterm birth, FGR, VLBW, miscarriage, gestational hypertension (decidual vasculopathy, acute atherosclerosis, infarcts). Diffuse disease in 45% of 214; renal crisis in 3.
- **Subfertility.** Avoid hormonal (especially combined) contraception with pulmonary, cardiac or renal involvement; consider sterilization.

7. Vasculitis Syndromes

- Primary or secondary; immune-complex deposition in most. **ANCA** in small-vessel vasculitis (GPA, EGPA). Overall ↑ **preterm birth and preeclampsia.**

Table 62-7. Primary Vasculitis Syndromes

Syndrome
Polyarteritis nodosa
Wegener granulomatosis
Takayasu arteritis
Behçet disease
Churg-Strauss
Henoch-Schönlein
Giant cell arteritis

Vasculitis	Key features	Pregnancy points
Polyarteritis nodosa	Necrotizing vasculitis of small/medium arteries : myalgia, neuropathy, GI, hypertension, renal; 1/3 hepatitis B antigenemia ; biopsy; prednisone + cyclophosphamide (HBV-related: antivirals, steroids, plasma exchange)	12 cases: onset in pregnancy in 7 — rapidly fatal by 6 wk postpartum , diagnosed at autopsy in 6
Granulomatosis with polyangiitis (Wegener)	Small-vessel necrotizing granulomatous : sinus/nasal 90% , lung 85% , GN 75% , musculoskeletal 65%; usually >50 yr	↑ Preterm birth; subglottic stenosis in up to 1/4 → antepartum anesthesia consult ; steroids ± azathioprine, cyclosporine, IVIG; cyclophosphamide + prednisolone for severe late-2nd/3rd-trimester disease; rituximab appears safe
Takayasu arteritis ("pulseless disease")	Large vessels (aortic arch and branches); onset <40 yr (vs temporal arteritis >55); death from CHF or stroke; CT/MR angiography; steroids (not curative), bypass/angioplasty	Prognosis generally good; half have complications (hypertension, preeclampsia); 5% life-threatening CV events ; measure BP in the lower extremity ; abdominal aortic involvement worse; vaginal delivery + epidural
Behçet disease	~1 per 100,000 births	↑ Miscarriage, smaller babies, preeclampsia, preterm, VTE; not a contraindication
EGPA (Churg-Strauss)	Rare; necrotizing vasculitis (can involve heart)	IVIG response reported; postpartum relapses
Henoch-Schönlein purpura	Purpura, arthralgia, abdominal pain, GI bleeding; uncommon after childhood	Skin lesions 3/4, arthralgia 1/2; responds to steroids ; good outcomes

8. Inflammatory Myopathies

- Prevalence **14–32 per 100,000**. **Polymyositis, dermatomyositis, inclusion-body myositis** → progressive (asymmetrical) weakness.
- Dermatomyositis = characteristic rash** with/before weakness. ↑ Muscle enzymes, abnormal EMG; **biopsy confirms** (perivascular/perimysial infiltrates).
- ⚠ **~15% of adults with dermatomyositis have a malignancy** (breast, lung, stomach, **ovary**); may be years apart.
- Treatment: **high-dose corticosteroids**, azathioprine or methotrexate, IVIG.
- Pregnancy**: 17 gravidas: hypertension 23%, antepartum hemorrhage 11%, **cesarean 88%**, preterm 35%. In 98 women, **80% had no disease worsening**. Outcome tracks disease activity; **new-onset disease in pregnancy is aggressive**.

9. Hereditary Connective Tissue Disorders

- Mutations in structural proteins (fibrillar **collagen types I, II, III**, elastin, proteoglycans). Main pregnancy concern: **aortic aneurysm/dissection**.

Disorder	Genetics / prevalence	Pregnancy risks and management
Marfan syndrome	Autosomal dominant; 1 in 3000–5000 ; both sexes; FBN1 (fibrillin) on 15q21 , high mutation rate; PGT/prenatal diagnosis possible in the 80% with a known mutation	Aortic media degeneration → aortic dilation/dissection, more common in pregnancy ; Marfan underlay 65% of aortic dissections in 27 pregnant women (see Ch 52)
Ehlers-Danlos syndrome	7 per 100,000 ; AD, AR or X-linked; types I–III AD (~30% each); collagen/procollagen defects	PPROM, preterm birth, APH/PPH, previa, cervical insufficiency, uterine rupture ; fragile tissue complicates repair and cesarean; type IV (vascular): great-vessel and uterine rupture ; iliac artery rupture deaths; neonatal skull fractures (type VIIC)
Osteogenesis imperfecta	Type I 1 in 20,000 , type II 1 in 60,000 ; up to 15 subtypes; AD, AR, sporadic; type I = COL1A1, mildest; type II lethal in utero	Blue sclerae, hearing loss, fractures (often 20–30 before pregnancy), dental defects; risks: fractures, scoliosis/restrictive lung disease, micrognathia, unstable C-spine, uterine rupture, CPD ; ↑ APH, abruption, FGR, malformations, preterm; fetus may fracture in utero or at delivery; bisphosphonates considered

Quick-Recall: Disease Activity in Pregnancy and Key Risks

Disorder	Effect of pregnancy	Key fetal/obstetric risk	Must-know management
SLE	⅓ better / ⅓ same / ⅓ worse	Stillbirth RR 16.5 , preeclampsia, FGR, neonatal lupus, heart block	Quiescent ≥6 mo; continue hydroxychloroquine ; aspirin; C3/C4 + dsDNA; growth scans q4 wk from 24 wk
Lupus nephritis	Flare 30%	Preeclampsia 27%, preterm ⅓	Tacrolimus, azathioprine; no mycophenolate/cyclophosphamide/ACEI
Anti-Ro/La	—	Heart block 2–3% (20% recurrence) ; rash 4–6 wk	Fetal echo/PR monitoring 18–26 wk ; hydroxychloroquine
APS	—	Stillbirth, early preeclampsia, FGR; triple-positive worst	Aspirin + heparin ; prior thrombosis → anticoagulate + 6 wk postpartum; no steroids for primary APS
Rheumatoid arthritis	Improves up to 70%	Preterm, FGR, hypertension	Hydroxychloroquine, sulfasalazine, prednisone, TNF inhibitors; no methotrexate/leflunomide ; flare postpartum 40–50%
Systemic sclerosis	Usually stable; reflux worse	Preterm, FGR	Renal crisis doesn't resolve with delivery; neuraxial anesthesia; PAH contraindicates pregnancy
Takayasu	Stable	Hypertension/preeclampsia	BP in legs; vaginal delivery + epidural
Polyarteritis nodosa	Often fatal if onset in pregnancy	—	Steroids + cyclophosphamide
Marfan	Aortic dilation ↑	Dissection	See Ch 52
Ehlers-Danlos (type IV)	—	Vessel and uterine rupture, PPRM	Careful tissue handling

Key Numbers to Memorize

- Unrecognized rheumatic disease **0.3%** → preeclampsia/FGR **25%**
- SLE: **90%** women; **1 in 500** reproductive-age women; **1 in 900** births; twin concordance **25% vs 2%**; ANA **84–98%**, anti-dsDNA **62–70%**, anti-Sm **25–38%**, anti-Ro **30–49%**, antiphospholipid **21–50%**, antihistone **70%**
- SLE diagnosis: **≥4 criteria**; maternal mortality **325/100,000**; stillbirth RR **16.5**; preeclampsia RR **3.4**
- Lupus nephritis: flare **30%**, maternal mortality **1.3%**; chronic hypertension **up to 30%**

- Hydroxychloroquine: SGA ↓ **85%**, heart block ↓ **50%**; prednisone **1–2 mg/kg**; methylprednisolone **1000 mg × 3 d**; azathioprine **2 mg/kg**; stress hydrocortisone **100 mg q8h × 48 h**
- Neonatal lupus rash **30–40%** at **4–6 wk**; recurrence **25%**; heart block **2–3%** → **20%**; mortality **~20%**; monitor **18–26 wk**
- SLE survival **95 / 90 / 78%** at 5 / 10 / 20 years
- APS: live births with treatment **>70%**; failure despite treatment **~30%**; recurrent loss **20–30%**; APAs in **30%** of SLE and **~5%** of obstetric population; LAC alone in only **20%** of APS; confirm tests **12 wk apart**; ≥2 APA types → failure **93%** vs **63%**
- Aspirin **60–80 mg**; UFH **5000–10,000 U q12h**; enoxaparin **40 mg daily**; IVIG **2 g/kg**; anticoagulate **6 wk** postpartum; triple positive live birth **30%**
- RA: prevalence **0.5–1%**, F:M **3:1**; improves **up to 70%**; postpartum flare **40–50%**; new onset **6×** in first 3 postpartum months; prednisone **7.5–20 mg**
- Scleroderma: F:M **5:1**; **1 in 22,000** pregnancies; ANA and Raynaud **95%**; pulmonary hypertension **15%**; renal crisis **up to 1/4**
- GPA: sinus **90%**, lung **85%**, GN **75%**; subglottic stenosis **1/4**. PAN: hepatitis B **1/3**. Takayasu onset **<40**
- Dermatomyositis malignancy **15%**; myopathy prevalence **14–32/100,000**
- Marfan **1 in 3000–5000** (FBN1, 15q21); EDS **7/100,000**; OI type I **1 in 20,000**, type II **1 in 60,000**